An extension to TLS and DTLS allows certificates to be replaced with *raw public keys* (https://tools.ietf.org/html/rfc7250). This allows the complex certificate parsing and validation code to be eliminated, along with support for many signature algorithms, resulting in a large reduction in code size. The downside is that keys must instead be manually distributed to all devices, but this can be a viable approach in some environments. Another alternative is to use *pre-shared keys*, which you'll learn more about in section 12.2.

> **DEFINITION *Raw public keys*** can be used to eliminate the complex code required to parse and verify X.509 certificates and verify signatures over those certificates. A raw public key must be manually distributed to devices over a secure channel, for example during manufacture.

The situation is somewhat better when you look at the symmetric cryptography used to secure application data after the TLS handshake and key exchange has completed. There are two alternative cryptographic algorithms that can be used instead of the usual AES-GCM and AES-CBC modes:

- Cipher suites based on AES in *CCM mode* provide authenticated encryption using only an AES encryption circuit, providing a reduction in code size compared to other AES modes. CCM is widely used in IoT applications for this reason.
- The *ChaCha20-Poly1305* cipher suites can be implemented securely in software with relatively little code and good performance on a range of CPU